

RePlastix Innovations: Transforming Plastic Waste into Sustainable Solutions

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Project Platform	Salesforce Developer Edition
Project Domain	Plastic Waste Management and Recycled-Product Operations

Project Overview

RePlastix Innovations is implemented as a Salesforce application for organizing plastic-waste collection, recycling-center information, recycled-product inventory, customer orders, and replenishment activities. The solution combines structured data management with validation, role-based security, Flow automation, and Apex-based transaction processing.

Implementation Areas

Area	Covered in this report
Data Management	Objects, tabs, application navigation, fields and relationships
Data Security	Roles, profiles, users and record-level access
Data Configuration	Validation, controlled values and inventory status logic
Automation	Scheduled Flow and Apex inventory processing
Verification	Functional records, notification and successful test coverage

MOTIVATION, OBJECTIVES AND USE CASE

Motivation

Plastic waste management involves several connected activities: material is collected, directed to a recycling center, converted into usable products, placed into inventory, sold through customer orders, and replenished when demand reduces available stock. When these activities are handled through disconnected records or manual follow-up, it becomes difficult to maintain a consistent view of operations.

The Salesforce implementation addresses this coordination requirement by placing the operational information inside one connected application. The project also demonstrates how declarative Salesforce features and Apex can work together: configuration handles structure and simple rules, while automation manages recurring checks and transaction-dependent updates.

Objectives

- Create a centralized Salesforce application for RePlastix Innovations.
- Model collection, recycling, product, order and replenishment information as connected records.
- Provide controlled fields and relationships so information is stored consistently.
- Apply validation rules that prevent invalid business data from being saved.
- Display product stock condition through a formula-driven inventory status.
- Automate low-stock follow-up through Scheduled Flow.
- Process order and approved-restock inventory changes through Apex.
- Support role-based access and record visibility according to organizational responsibility.
- Verify the implementation through functional records and Apex test execution.

Use Case

Component	Use Case	System Outcome
Collection User	Register plastic waste	Collection record is stored with type, weight, date and center
Operations User	Maintain recycling-center information	Center records provide location and capacity context
Warehouse / Sales User	Create customer order	Order references a recycled product and quantity
Inventory Process	Check available stock	Stock is reduced when sufficient quantity exists
Inventory Process	Handle shortage	A replenishment request is created for unavailable quantity
Authorized User	Approve replenishment	Product stock is increased and notification is issued
Scheduled Flow	Check low stock	A Task is created when a product falls below threshold

Expected End-to-End Outcome

The intended sequence is collection → recycling-center association → recycled-product inventory → customer order → stock evaluation → replenishment request when required → approval → inventory update and notification. The same application provides the records and navigation needed to inspect each stage.

SALESFORCE CREDENTIALS SETUP

3.1 Salesforce Developer Environment

The project was implemented in a Salesforce Developer Edition environment. The Developer Edition org provides the required workspace for creating custom objects, fields, Lightning application components, security configuration, Flow automation and Apex classes. The implementation was performed through Salesforce Setup and the Lightning application interface.

3.2 Secure Login Practice

Salesforce credentials are private account information and are not reproduced in this report. For project execution, the Salesforce login was used to access the Developer Edition org, while all configuration work was carried out inside the authorized project environment.

Setup Item	Configuration Approach
Salesforce Org	Developer Edition environment used for project development
Authentication	Salesforce login through the authorized project account
Development Access	Setup, Object Manager, App Manager, Flow Builder and Apex tools
Application Access	Re Plastic Innovations Lightning application
Testing Access	Developer Console and Apex test execution
Credential Handling	Credentials retained privately; no passwords or secrets recorded

3.3 Initial Workspace Preparation

- Sign in to the Salesforce Developer Edition environment.
- Open Setup to access configuration and metadata management.
- Confirm that Object Manager, App Manager, Flow Builder and security configuration are available.
- Use the project application for day-to-day record-level verification.
- Use Developer Console only for Apex execution and testing.
- Keep account credentials outside the project report and screenshots.

3.4 Implementation Principle

The credentials stage is deliberately treated as an access foundation rather than a functional project feature. Once the authorized Salesforce environment was available, the work could proceed systematically through data management, security, configuration and automation. No account password or authentication secret is required for understanding or reproducing the technical design described in the remaining pages.

3.5 Environment Readiness

A ready Salesforce environment provides two complementary areas: Setup for defining the application and its metadata, and the Lightning workspace for using the resulting records. The project uses both areas throughout the implementation, with each configuration step later verified through the user-facing application wherever appropriate.

DATA MANAGEMENT – OBJECTS

4.1 Object Model

The data model was designed around five business entities. Each object represents one clear responsibility so that collection information, processing locations, product inventory, customer demand and replenishment activity remain logically separated. Lookup relationships then connect the records where the business process requires navigation between them.

Custom Object	Purpose	Key Information
Plastic Waste	Records collected material	Collection Date, Type, Weight, Status, Location, Recycling Center
Recycling Center	Represents processing facilities	Name, Location, Capacity
Recycled Product	Maintains finished-product inventory	Product Name, Stock Level, Threshold, Price, Stock Status
Order	Captures customer demand	Customer, Recycled Product, Quantity, Delivery Date
Restock Request	Tracks replenishment needs	Product, Request Quantity, Status

4.2 Relationship Structure

Plastic Waste is associated with a Recycling Center. Order references Recycled Product so that demand can be matched to inventory. Restock Request also references Recycled Product, allowing a shortage and its later replenishment to remain tied to the same inventory record.

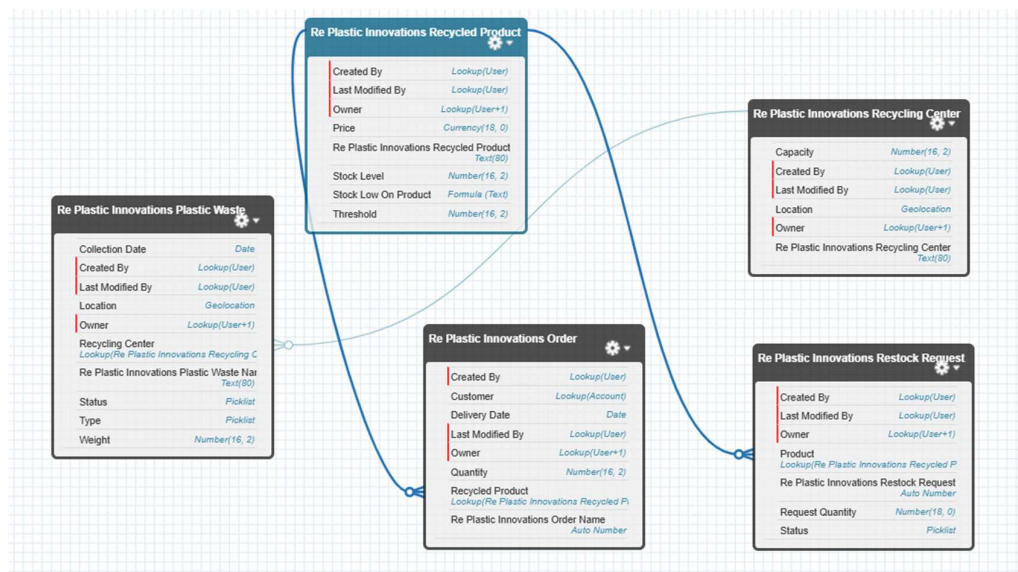


Figure 1. Connected custom-object structure created for the RePlastix Innovations application.

4.3 Why Separate Objects Were Used

- Each record type has a focused business purpose.
- Repeated information is avoided by using relationships instead of duplicating details.
- Inventory fields remain on Recycled Product, where they can be reused by Flow and Apex.
- Order records describe demand, while Restock Request records describe replenishment.
- The structure supports clearer navigation, security and reporting.

4.4 Object Creation Sequence

The objects were established before automation because every later component depends on stable object and field references. Once the object layer was available, tabs, fields, relationships, validation rules and automation could be configured against a defined data model.

DATA MANAGEMENT – TABS

5.1 Creating Accessible Object Tabs

Custom tabs provide the direct navigation layer for the project objects. Instead of requiring users to enter Salesforce Setup, the application exposes the records through dedicated navigation items. This makes the project easier to operate and also provides a clear connection between the data model and the user interface.

Tab	Navigation Purpose
Plastic Wastes	Open and manage collection records
Recycling Centers	Maintain recycling-location records
Recycled Products	Review inventory and product status
Orders	Create and inspect customer orders
Restock Requests	Review replenishment records and approval status

5.2 Tab Configuration Process

- Open Setup and navigate to the custom object configuration.
- Ensure each required custom object has a corresponding tab.
- Use clear plural navigation labels for list-oriented records.
- Expose the tabs to the Re Plastic Innovations application.
- Arrange the navigation items in an order that follows the operational workflow.
- Open the Lightning application and confirm that the navigation items are visible.

5.3 Navigation Evidence

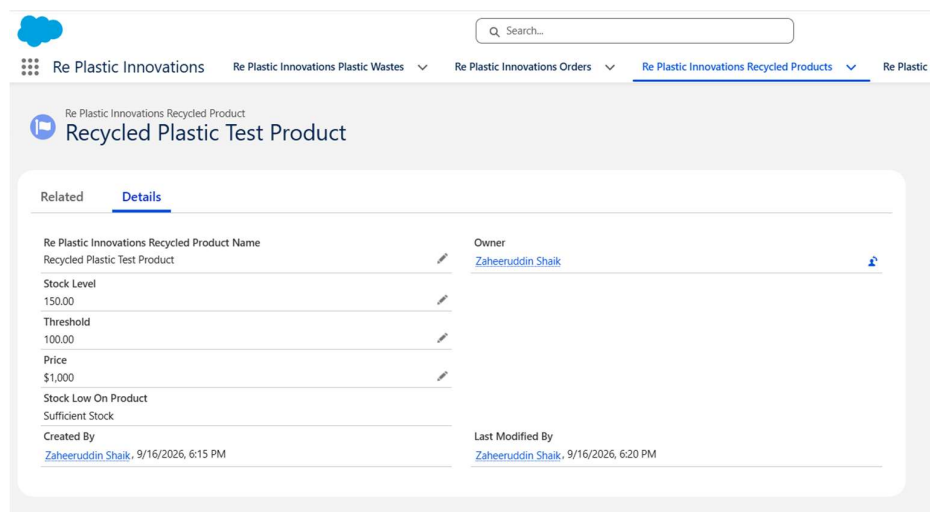


Figure 2. Re Plastic Innovations navigation bar showing the project record areas, including Recycled Products.

The application navigation visible in the supplied record screenshot demonstrates that the custom business areas are exposed through the Lightning interface. The navigation sequence supports movement from operational records to inventory and replenishment information.

5.4 Usability Consideration

The tabs were treated as part of the application design rather than merely as technical metadata. Clear names reduce the time needed to locate a record and make demonstrations easier.

DATA MANAGEMENT – APP MANAGER

6.1 Re Plastic Innovations Lightning Application

The Re Plastic Innovations application was configured as the main user workspace. App Manager provides the place where the navigation items, branding context and application-level organization are brought together. The objective was to give the project a recognizable Salesforce interface instead of presenting the custom objects as isolated Setup components.

6.2 Application Configuration Steps

- Open App Manager from Salesforce Setup.
- Create or configure the Re Plastic Innovations Lightning application.
- Set the application name so it represents the project domain.
- Add the required custom object tabs to the navigation.
- Place frequently used operational objects in a practical order.
- Save the application configuration.
- Open the application from the App Launcher and verify the navigation.

6.3 Recommended Navigation Order

Position	Application Item	Reason
1	Plastic Wastes	Entry point for collection information
2	Orders	Captures customer demand
3	Recycled Products	Central inventory reference
4	Restock Requests	Supports replenishment monitoring
5	Recycling Centers	Provides processing-location information

6.4 Application-Level Experience

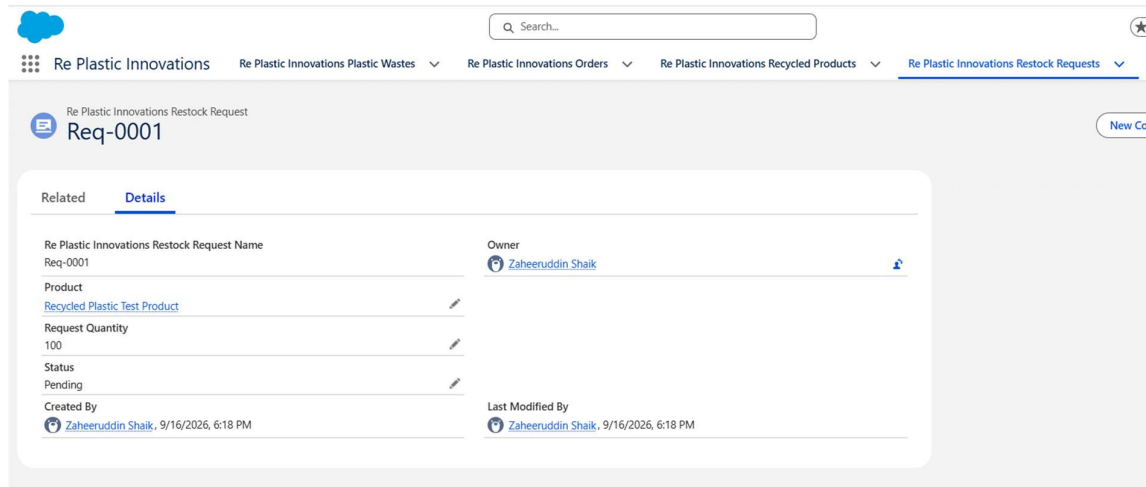


Figure 3. Re Plastic Innovations application displaying a Restock Request record through the configured workspace.

The record page shows the application navigation across the top and a dedicated Re Plastic Innovations Restock Request record below it. This demonstrates the practical result of App Manager configuration: project records can be accessed directly from a common Lightning workspace.

6.5 Outcome

App Manager therefore acts as the bridge between Salesforce configuration and daily project usage. The object tabs created earlier become meaningful to users only after they are placed into an application where the operational sequence is easy to follow.

DATA MANAGEMENT – FIELDS

7.1 Field Design Strategy

Fields were created according to the information each business entity must retain. The implementation uses dates for time-based events, numbers for measurable quantities, picklists for controlled states, geolocation for collection or facility location, currency for product price, and lookups for links between related objects.

7.2 Plastic Waste Fields

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Collection Date	Collection_Date__c	Date		
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Location	Location__c	Geolocation		
Owner	OwnerId	Lookup(User,Group)		✓
Re Plastic Innovations Plastic Waste Name	Name	Text(80)		✓
Recycling Center	Recycling_Center__c	Lookup(Re Plastic Innovations Recycling Center)		✓
Status	Status__c	Picklist		
Type	Type__c	Picklist		
Weight	Weight__c	Number(16, 2)		

Figure 4. Plastic Waste Fields & Relationships configuration.

Field	Type	Purpose
Collection Date	Date	Records the date associated with collection
Location	Geolocation	Stores the collection location
Recycling Center	Lookup	Associates waste with a processing center
Status	Picklist	Represents the collection/processing state
Type	Picklist	Controls the material category
Weight	Number	Stores the collected quantity

7.3 Field Design Principle

The screenshot confirms that the Plastic Waste object contains a structured field set rather than a single free-form description. This structure allows the same information to be used consistently by validation, reporting and related-record navigation.

7.4 Relationship Fields

The Recycling Center lookup is particularly important because it connects collection records to the facility responsible for processing them. Relationships preserve the logical flow of the application without repeating facility information on every waste record.

DATA MANAGEMENT – FIELDS: INVENTORY AND CONTROLLED VALUES

8.1 Recycled Product Fields

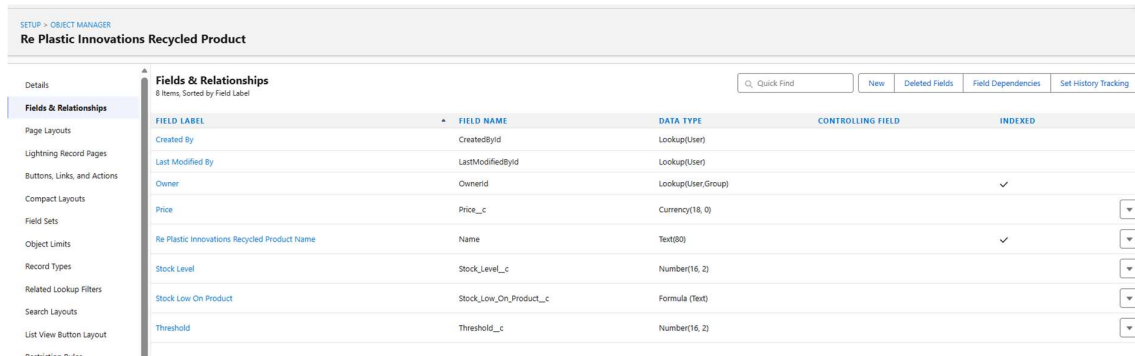


Figure 5. Recycled Product Fields & Relationships configuration.

Field	Data Type	Role in Inventory
Recycled Product Name	Text	Identifies the product
Stock Level	Number	Stores currently available stock
Threshold	Number	Defines the low-stock boundary
Price	Currency	Stores the product value
Stock Low On Product	Formula (Text)	Presents a readable stock condition

8.2 Controlled Material Categories

The Plastic Waste Type field uses controlled picklist values. This prevents inconsistent naming of material categories and gives users a predictable set of choices when creating collection records.

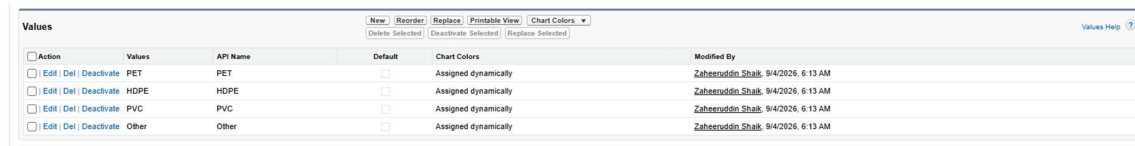


Figure 6. Configured Plastic Waste Type values: PET, HDPE, PVC and Other.

8.3 Value Standardization

- PET represents polyethylene terephthalate material.
- HDPE represents high-density polyethylene material.
- PVC represents polyvinyl chloride material.
- Other provides a controlled category for materials outside the primary choices.

Using a picklist turns the Type field into a controlled data-entry point. This improves consistency for filtering, grouping and future reporting because the same category is represented by the same stored value across records.

8.4 Field-Level Outcome

The combined field configuration establishes the information needed by the later business rules. Stock Level and Threshold support inventory evaluation, while Plastic Waste fields provide the structured collection data needed for the project's operational records.

DATA SECURITY – ROLES

9.1 Role Hierarchy

The role hierarchy was configured to represent organizational responsibility within the RePlastix Innovations environment. Roles provide a record-visibility structure that can be used alongside profiles and sharing settings. The hierarchy separates management oversight from operational responsibilities.

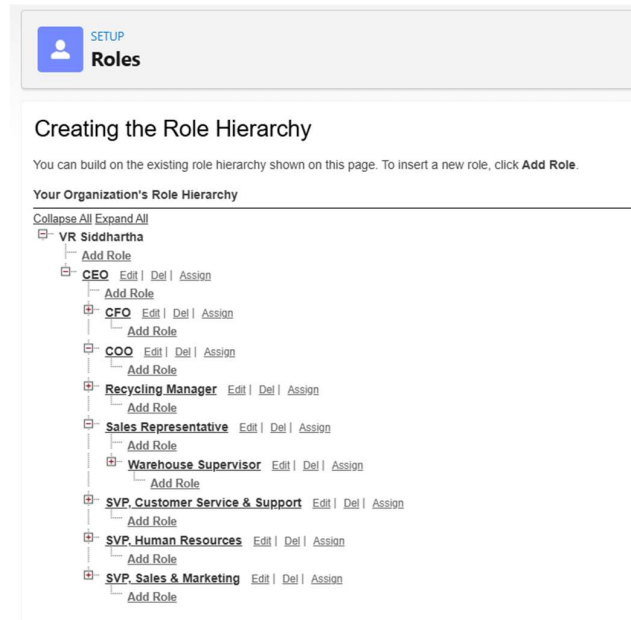


Figure 7. Salesforce role hierarchy configured in the project environment.

9.2 Roles Used for the Project

Role	Primary Responsibility
CEO	Overall organizational oversight
Recycling Manager	Plastic-waste and recycling activities
Sales Representative	Customer orders and product demand
Warehouse Supervisor	Inventory and replenishment activities

9.3 Configuration Sequence

- Open Setup and select Roles.
- Review the organization root and existing hierarchy.
- Create project-specific roles where operational responsibility requires them.
- Place roles beneath the appropriate management level.
- Assign users to their corresponding roles after the hierarchy is established.
- Use profiles and sharing configuration separately to define functional access.

9.4 Security Outcome

The role hierarchy provides an organizational context for record access. It is intentionally complemented by profiles, users and record-level security so that access decisions are based on both responsibility and permissions rather than on the hierarchy alone.

DATA SECURITY – PROFILES

10.1 Purpose of Profiles

Profiles define the baseline permissions available to Salesforce users. In this project, profiles are used to control which objects and functions can be accessed by different categories of users. This complements the role hierarchy, which addresses organizational position and record visibility.

10.2 Profile Configuration Approach

Profile Area	Configuration Consideration
Object Permissions	Grant the required Create, Read, Edit and Delete access for project objects
Field Permissions	Expose only the fields required for the user's responsibilities
Application Access	Allow appropriate users to open the Re Plastic Innovations application
Tab Visibility	Make relevant project tabs available to the profile
System Permissions	Keep administrative capabilities limited to authorized users

10.3 Implementation Steps

- Open Setup and navigate to Profiles.
- Select the profile associated with the intended operational user group.
- Review object permissions for Plastic Waste, Recycling Center, Recycled Product, Order and Restock Request.
- Set tab visibility so the profile sees the required project navigation items.
- Review field-level access for sensitive or operationally specific fields.
- Save the configuration and verify access with the assigned user.

10.4 Principle of Least Required Access

The project follows a least-required-access approach: users should receive the capabilities needed for their work instead of broad administrative access. Profiles establish the baseline, while permission sets can be used when an additional capability is required without changing the entire profile.

10.5 Relationship with Roles

A role answers where the user sits in the organizational structure; a profile answers what the user can do. Keeping these responsibilities separate makes the security model easier to understand and maintain.

DATA SECURITY – USERS

11.1 User Assignment

After roles and profiles are established, Salesforce users can be associated with the appropriate combination. The user record connects an individual login to a profile, role and application access. This provides the identity layer through which the security model becomes operational.

11.2 User Configuration Sequence

- Open Setup and navigate to Users.
- Create or open the project user record.
- Assign the appropriate profile according to job responsibility.
- Assign the appropriate role from the configured role hierarchy.
- Confirm that the Re Plastic Innovations application is available.
- Save the user configuration.
- Verify the user's access from the Lightning application.

11.3 User-to-Responsibility Mapping

Operational Area	Suitable User Responsibility	Related Project Objects
Collection	Recycling Manager	Plastic Waste, Recycling Center
Sales	Sales Representative	Order, Recycled Product
Warehouse	Warehouse Supervisor	Recycled Product, Restock Request
Management	Executive Roles	Visibility across operational records

11.4 Access Verification

User verification should be performed by opening the application with the intended permissions and checking whether the expected tabs, records and actions are available. This confirms that profile permissions, role assignment and application visibility work together.

11.5 Security Outcome

The completed user layer turns the abstract security design into an executable access model. Each user receives a defined organizational position and a baseline set of Salesforce permissions, while record-level rules provide the additional visibility control described on the next page.

DATA SECURITY – RECORD LEVEL SECURITY

12.1 Record Visibility Model

Record-level security determines which individual records a user can see or modify after object permissions have been granted. For the project, record access is considered separately from object access so that a user can have permission to work with an object without automatically receiving unrestricted visibility of every record.

12.2 Security Layers

Security Layer	Purpose in the Project
Organization-Wide Defaults	Establish the baseline record visibility
Role Hierarchy	Supports visibility through organizational responsibility
Sharing Rules	Provide additional access when business collaboration requires it
Manual / Ownership Access	Allows record ownership to participate in access control
Profiles / Permission Sets	Define object and functional permissions

12.3 Configuration Procedure

- Review organization-wide defaults for the project objects.
- Set the baseline visibility according to the intended operating model.
- Use the role hierarchy to extend visibility where managerial access is required.
- Create sharing rules when users in different branches need selected records.
- Confirm that object permissions are consistent with record-level visibility.
- Test the final arrangement using users representing different operational roles.

12.4 Relationship to the Role Screenshot

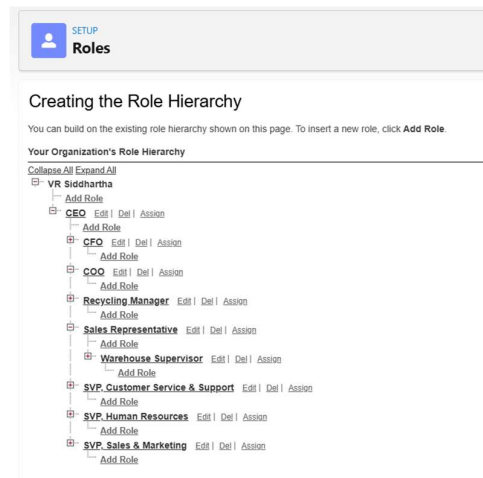


Figure 8. Role hierarchy used as one layer of the project's record-security model.

The role hierarchy shown above provides the organizational structure that record visibility can use. Profiles and record-level controls remain separate layers, allowing the application to distinguish between a user's functional permission and the specific records that user should access.

12.5 Security Outcome

This layered approach reduces the risk of treating security as a single setting. Object permissions, user assignments, roles and record-sharing behavior work together to provide controlled access to the RePlastix Innovations data.

DATA CONFIGURATION – INVENTORY STATUS

13.1 Stock Status Formula

The Recycled Product object contains a formula field named Stock Low On Product. Its purpose is to convert the numeric relationship between Stock Level and Threshold into a readable status. This gives users an immediate indication of whether the inventory requires attention.

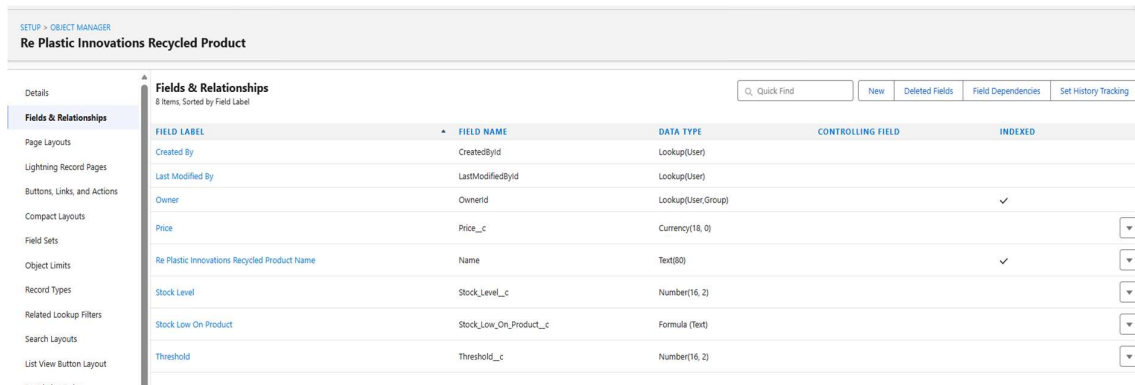


Figure 9. Recycled Product field configuration containing Stock Level, Threshold and the stock-status formula.

13.2 Formula Logic

Condition	Displayed Text	Interpretation
Stock Level < Threshold	Low Stock - Restock Needed	Inventory has crossed the defined low-stock boundary
Stock Level >= Threshold	Sufficient Stock	Inventory is at or above the configured threshold

13.3 Configuration Steps

- Open Object Manager and select Recycled Product.
- Open Fields & Relationships and create the stock-status formula field.
- Set the formula return type to Text.
- Compare Stock Level with Threshold.
- Return the low-stock message when Stock Level is below Threshold.
- Return the sufficient-stock message for the remaining condition.
- Save the field and verify the result from a Recycled Product record.

13.4 Record-Level Verification

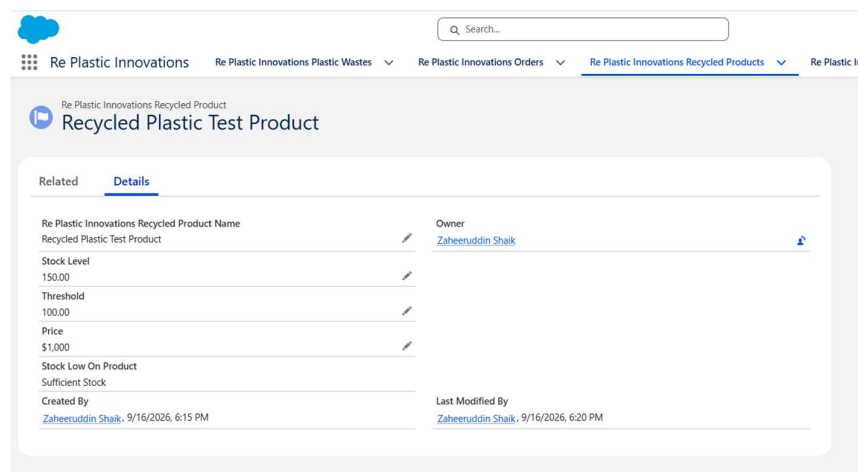


Figure 10. Recycled Plastic Test Product record showing Stock Level 150, Threshold 100 and the resulting Sufficient Stock status.

The supplied record demonstrates the formula behavior in the user-facing application. Because the Stock Level is 150 and the Threshold is 100, the displayed status is Sufficient Stock. This confirms that the formula is being evaluated from the record's stored values.

DATA CONFIGURATION – VALIDATION AND CONTROLLED INPUT

14.1 Collection-Date Validation

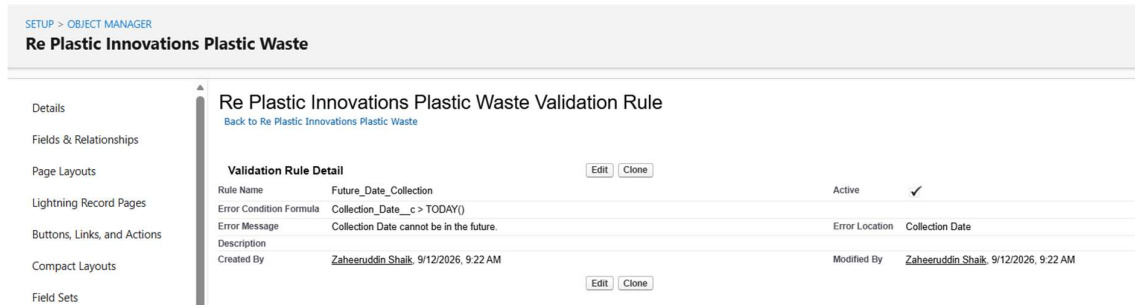


Figure 11. Active Plastic Waste validation rule for preventing future collection dates.

The Plastic Waste validation rule compares Collection Date with the current date. When the entered date is later than today, the record is not accepted. The configured message, “Collection Date cannot be in the future,” gives a direct explanation at the field where the invalid value is entered.

14.2 Order Quantity Validation

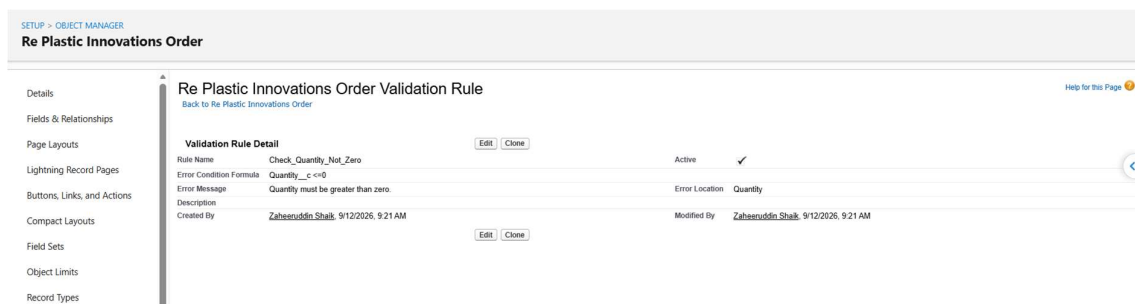


Figure 12. Active Order validation rule requiring Quantity to be greater than zero.

The Order object uses a validation rule that rejects Quantity values less than or equal to zero. This keeps the order transaction meaningful before it reaches the inventory-processing logic.

14.3 Configuration Procedure

- Open Object Manager and select the relevant object.
- Open Validation Rules and create the rule using a clear descriptive name.
- Define the condition that represents invalid data.
- Enter an explanatory error message.
- Set the error location to the related field.
- Activate the rule and save the configuration.
- Verify that valid values remain available for normal record creation.

14.4 Controlled Input and Validation Together

Picklists restrict categorical values, while validation rules enforce business conditions. Using both mechanisms creates a stronger data-entry layer: the user chooses from controlled categories and the record is prevented from violating the project's basic rules.

AUTOMATION – FLOW BUILDER

15.1 Scheduled Low-Stock Flow

The project uses Scheduled-Triggered Flow for recurring inventory monitoring. The Flow is configured to run daily against Recycled Product records and evaluate the low-stock condition. When a product falls below its threshold, the Flow creates a Task so that the record owner receives a concrete follow-up item.

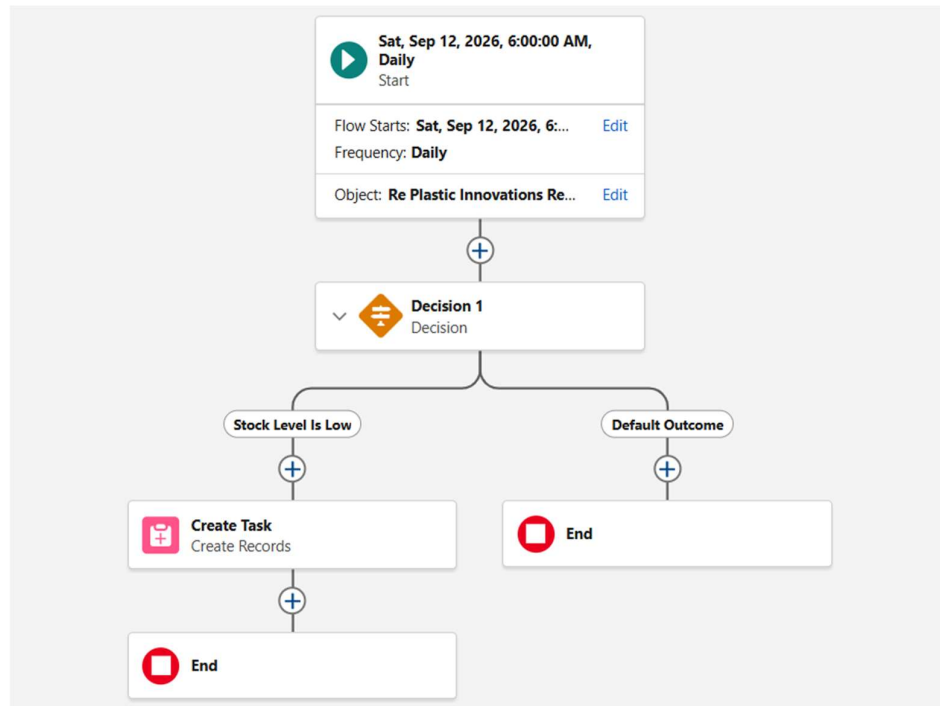


Figure 13. Scheduled Flow with a low-stock decision and Create Task path.

15.2 Flow Configuration

Flow Element	Configured Behavior
Start	Scheduled daily execution from 12 September 2026 at 6:00 AM
Object	Re Plastic Innovations Recycled Product
Decision 1	Checks the low-stock condition
Stock Level Is Low	Routes matching products to Task creation
Create Task	Creates a follow-up record
Default Outcome	Ends the path when the low-stock condition is not met

15.3 Execution Steps

- Create a Scheduled-Triggered Flow.
- Select Recycled Product as the object.
- Set the schedule and daily frequency.
- Add a Decision element for the low-stock condition.
- Create a Task on the low-stock branch.
- Associate the Task with the appropriate product and owner information.
- Save and activate the Flow.
- Review the Flow canvas to confirm that the decision and action paths are connected.

15.4 Why Flow Is Suitable

The requirement is time-based rather than dependent on a particular user transaction. Scheduled Flow therefore provides a declarative way to perform the check repeatedly without placing scheduling responsibility inside the Apex transaction layer.

AUTOMATION – APEX

16.1 Purpose of Apex Automation

Apex is used for the parts of the RePlastix Innovations process that require transaction-oriented processing. The Apex layer handles inventory decisions associated with orders and replenishment approvals. It works with the custom objects and their relationships to update product stock and create or process replenishment records.

16.2 Programmatic Responsibilities

Component	Responsibility
InventoryManager	Central inventory-processing logic for order and replenishment operations
UpdateStockAfterOrder	Connects new Order records to inventory processing
UpdateStockAfterRestockApproval	Connects newly approved Restock Request records to inventory processing
EmailNotificationHelper	Sends the notification after restock approval processing

16.3 Order Processing

- Identify the recycled product referenced by the order.
- Read the current product stock.
- Compare the available quantity with the order quantity.
- Reduce Stock Level when sufficient inventory is available.
- Create a replenishment request when the requested quantity cannot be fulfilled.
- Persist the required record changes within the Salesforce transaction.

16.4 Restock Approval Processing

- Detect a Restock Request whose status changes to Approved.
- Identify the related Recycled Product.
- Increase the product Stock Level by the approved replenishment quantity.
- Invoke the notification helper after successful processing.
- Keep the trigger focused on the business event while the central class performs the reusable logic.

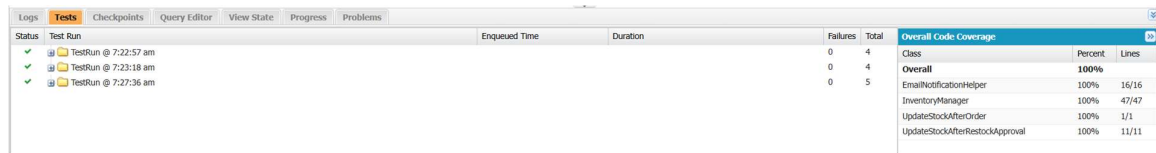
16.5 Architecture Principle

The project keeps the trigger layer focused and places reusable business processing in InventoryManager. This makes the Apex design easier to test and keeps transaction behavior separated from the event that starts it.

AUTOMATION – APEX: SUCCESSFUL TEST EXECUTION

17.1 Apex Test Verification

The completed Apex implementation was verified through Salesforce test execution. The supplied Developer Console result shows three successful test-run entries, each with zero failures. The coverage panel reports 100% overall coverage for the displayed project classes and triggers.



The screenshot shows the Salesforce Developer Console interface. The 'Tests' tab is active, displaying a table of test runs. All three test runs are successful (green checkmarks) and have zero failures. The 'Overall Code Coverage' panel on the right shows 100% coverage for all classes and triggers.

Status	Test Run	Enqueued Time	Duration	Failures	Total
✓	TestRun @ 7:22:57 am			0	4
✓	TestRun @ 7:23:18 am			0	4
✓	TestRun @ 7:27:36 am			0	5

Overall Code Coverage		
Class	Percent	Lines
Overall	100%	
EmailNotificationHelper	100%	16/16
InventoryManager	100%	47/47
UpdateStockAfterOrder	100%	1/1
UpdateStockAfterRestockApproval	100%	11/11

Figure 14. Developer Console test results showing zero failures and 100% overall code coverage.

17.2 Observed Coverage

Class / Trigger	Coverage
EmailNotificationHelper	100% (16/16 lines)
InventoryManager	100% (47/47 lines)
UpdateStockAfterOrder	100% (1/1 line)
UpdateStockAfterRestockApproval	100% (11/11 lines)
Overall	100%

17.3 Test Scenarios Represented

- Sufficient-stock order processing.
- Insufficient-stock handling and replenishment request creation.
- Approved restock processing and product-stock increase.
- Notification helper execution.
- Trigger and supporting-class execution through the Salesforce test framework.

17.4 Result Interpretation

The screenshot provides direct behavioral evidence for the Apex layer: the recorded test runs have zero failures, and every displayed Apex component has full line coverage. This is stronger evidence than simply showing that the classes exist because the test execution confirms that the implementation paths were exercised successfully.

AUTOMATION – APEX: RESTOCK APPROVAL AND NOTIFICATION

18.1 Restock Request Record

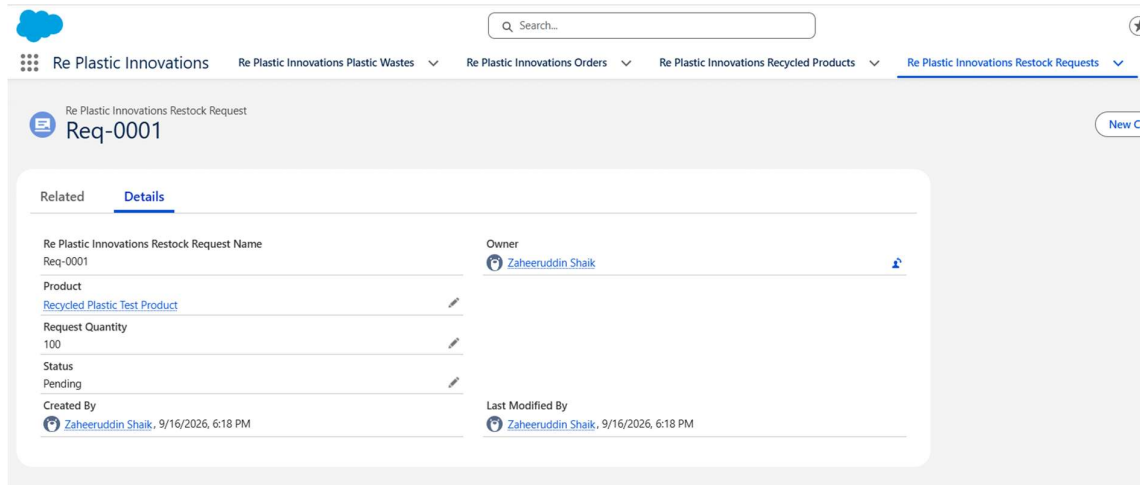


Figure 15. RePlastix Innovations Restock Request record displayed in the Salesforce application.

The Restock Request record provides the operational record for replenishment. It links the request to the Recycled Plastic Test Product, stores the requested quantity and exposes the current status. The record is therefore the bridge between an inventory shortage and the later approval-driven stock update.

18.2 Approval-Based Processing

- A shortage is represented by a Restock Request.
- The request carries the product relationship and replenishment quantity.
- The request status is changed to Approved by an authorized user.
- The approval trigger invokes the inventory-processing layer.
- The related product stock is increased by the approved quantity.
- The notification helper communicates the approval event.

18.3 Notification Evidence

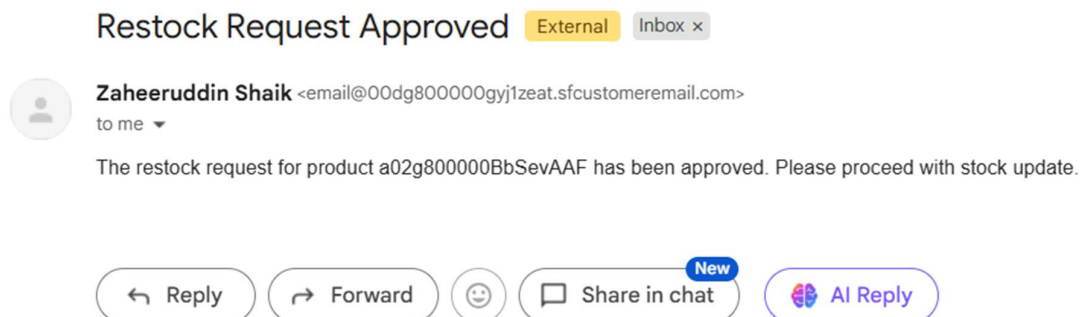


Figure 16. Restock Request Approved email notification captured from the completed implementation.

The supplied email confirms that the approval stage produces a user-facing notification. The message identifies the restock request as approved and instructs the recipient to proceed with the stock update, demonstrating the communication part of the automation.

18.4 Business Value

This automation removes the need for a user to manually communicate every approval. The Salesforce record remains the source of process state, Apex performs the inventory action, and the email provides immediate confirmation to the responsible operational user.

INTEGRATED DATA FLOW AND FUNCTIONAL VERIFICATION

19.1 From Collection to Inventory

The completed data model begins with Plastic Waste and Recycling Center records. Collection information is structured through fields and controlled values, while the lookup relationship connects the material to the appropriate processing center. Recycled Product then represents the usable output and its available stock.

19.2 From Inventory to Customer Order

Customer orders reference a Recycled Product record and carry a positive quantity. The order validation rule establishes the minimum data condition, while Apex evaluates the actual stock relationship. This creates a clear separation between data acceptance and transaction processing.

19.3 From Shortage to Restock

When available stock cannot satisfy demand, the replenishment process provides a separate Restock Request record. Approval of that request becomes the event that permits inventory replenishment. The product record then remains the central source for current stock information.

19.4 Final Product Record Evidence

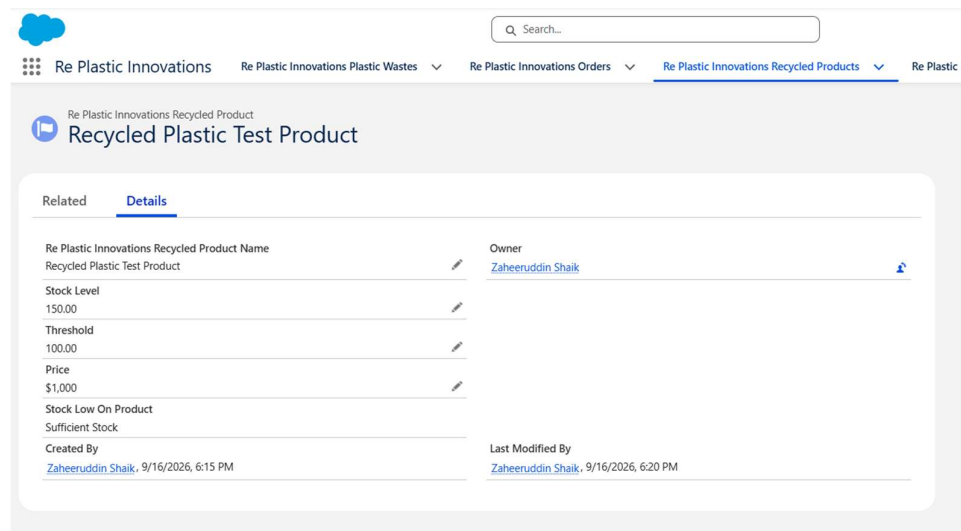


Figure 17. Recycled Plastic Test Product record used for functional verification.

Observed Record Value	Functional Meaning
Stock Level: 150.00	Current inventory quantity
Threshold: 100.00	Low-stock boundary
Price: \$1,000	Configured product value
Stock Low On Product: Sufficient Stock	Formula result for the displayed values
Owner: Zaheeruddin Shaik	Record owner available for operational assignment

19.5 Integrated Outcome

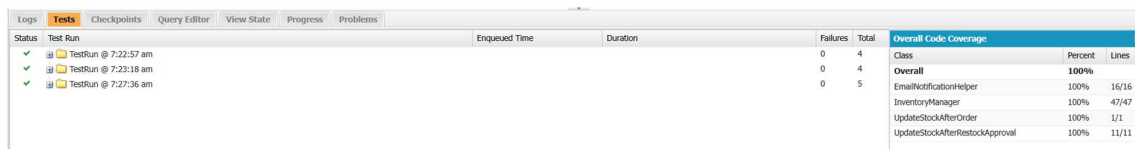
The record view confirms that the custom fields, formula result and ownership information are visible together in the Lightning interface. This provides a practical validation point for the data-management and automation layers described in the preceding pages.

FINAL IMPLEMENTATION SUMMARY AND CONCLUSION

20.1 Completed Implementation Sequence

Stage	Completed Project Result
Salesforce Credentials Setup	Authorized Developer Edition environment prepared
Objects	Five connected custom business entities created
Tabs	Project objects exposed through navigation
App Manager	Re Plastic Innovations Lightning application configured
Fields	Operational, numeric, date, location, lookup and controlled fields configured
Roles	Organizational hierarchy established
Profiles	Baseline functional permissions defined
Users	User-to-role and profile assignment supported
Record-Level Security	Layered record visibility model defined
Data Configuration	Formula, picklists and validation rules implemented
Flow Builder	Daily low-stock monitoring and Task creation configured
Apex	Inventory processing, restock approval and notification implemented

20.2 Final Evidence



Status	Test Run	Enqueued Time	Duration	Failures	Total
✓	TestRun @ 7:22:57 am			0	4
✓	TestRun @ 7:23:18 am			0	4
✓	TestRun @ 7:27:36 am			0	5

Overall Code Coverage:		
Class	Percent	Lines
Overall	100%	
EmailNotificationHelper	100%	16/16
InventoryManager	100%	47/47
UpdateStockAfterOrder	100%	1/1
UpdateStockAfterRestockApproval	100%	11/11

Figure 18. Final Apex verification showing zero failures and 100% overall code coverage.

20.3 Conclusion

The RePlastix Innovations project demonstrates a complete Salesforce-based operational solution in which data management, security, configuration and automation are connected rather than implemented as isolated features. The five-object data model provides a clear representation of the business process, while tabs and App Manager make that model accessible through a dedicated Lightning application.

The security layer establishes organizational responsibility and controlled access. Data configuration improves reliability through picklists, validation rules and a stock-status formula. Scheduled Flow provides recurring low-stock follow-up, while Apex handles transaction-based inventory processing and approved replenishment. The supplied test evidence records zero failures and 100% overall coverage for the displayed Apex components.

20.4 Final Project Outcome

- Centralized management of plastic-waste and recycled-product information.
- Connected inventory, order and replenishment records.
- Controlled and validated business data.
- Role-oriented access and record visibility.
- Automated low-stock follow-up.
- Programmatic stock processing for orders and approved restocks.
- Email notification for approved replenishment.
- Successful Apex test execution with 100% displayed code coverage.